# Vision-Language Models for Interpretable Chest X-Ray Diagnosis Using the MIMIC-CXR Dataset

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## Abstract

Vision–language models (VLMs) have emerged as powerful tools for automated chest X-ray interpretation, offering improvements in diagnostic accuracy, multimodal reasoning, and clinical workflow support. However, their adoption in high-stakes medical settings remains limited by concerns around transparency and explainability. This literature review examines recent advances in medical VLMs, focusing on models such as CXR-CLIP, CheXagent, and BioViL-T, and evaluates how architectural design choices influence multimodal alignment, temporal reasoning, and interpretability. Particular attention is given to BioViL-T due to its clinically meaningful spatio-temporal attention mechanisms and its suitability for anatomically grounded explainability. Based on these insights, the project proposes a comprehensive plan to compare BioViL-T and CXR-CLIP using a curated subset of the MIMIC-CXR-JPG dataset, implementing a wide range of interpretability techniques which incorporate attention rollout, gradient-based attribution, concept-based methods, and counterfactual analyses. The planned evaluation aims to identify strengths, limitations, and potential failure modes in the interpretability of modern VLMs for chest X-ray diagnosis.

☑ **I certify that all material in this dissertation which is not my own has been identified.**

**Signature: Muhammad Taha Mukati**

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# 1 Introduction

Chest X-rays (CXRs) remain the most frequently performed diagnostic imaging examination worldwide, with over 70 million conducted annually in the United States alone [1], serving as a cost-effective and non-invasive means of detecting thoracic conditions such as pneumonia, tuberculosis, and cardiomegaly [2]. However, interpreting these images is inherently challenging due to subtle radiographic findings, inter-observer variability, and overwhelming caseloads that strain clinical resources [3]. Over the past decade, deep learning has offered a pathway to alleviate these pressures, with Convolutional Neural Networks (CNNs) achieving radiologist-level performance in tasks such as pneumonia detection [4]. Yet these vision-only approaches cannot capture the rich contextual information embedded in clinical narratives, driving the emergence of vision-language models (VLMs) that integrate visual and textual data for more comprehensive diagnostic support [5].

Despite their diagnostic promise, VLMs remain largely opaque, offering limited insight into how they arrive at their predictions [6]. This lack of interpretability poses a critical barrier to clinical adoption, as clinicians and healthcare regulators increasingly require transparent and explainable reasoning before AI-driven tools can be trusted and deployed in practice [3, 7]. While a growing body of research has explored individual interpretability techniques in isolation, no prior work has systematically compared multiple complementary methods across architecturally distinct VLMs within a clinical radiology setting. This project directly addresses that gap by applying and evaluating a variety of interpretability approaches to two leading biomedical VLMs, BioViL-T and CXR-CLIP, using a curated subset of the MIMIC-CXR-JPG dataset. Through this systematic comparison, the project aims to reveal how architectural differences between the two models influence the clarity, reliability, and clinical relevance of their explanations, ultimately informing more principled decisions about how interpretability methods should be selected and applied in medical imaging AI.

# 2 Background

## 2.1 Motivation

The clinical need for automated chest X-ray interpretation tools is intensifying. Imaging volumes continue to grow by approximately 5% per year, while radiology residency positions increase by only 2%, creating a widening gap between demand and work force capacity [8]. In the UK, the radiology workforce faced a shortfall of 33%, with direct consequences for patient safety and service delivery [9]. While deep learning models offer a promising response, their clinical adoption hinges on whether clinicians can understand and trust their outputs. This trust deficit is not hypothetical. IBM Watson for Oncology, despite its technical sophistication, saw limited uptake precisely because its opaque reasoning left oncologists unable to verify its recommendations [10]. Regulatory frameworks are also moving to address this gap: the EU AI Act mandates transparency for high-risk systems, and the FDA's Good Machine Learning Practice framework similarly requires explainable outputs [11]. These developments make interpretability not only a research concern but a prerequisite for deployment.

Crucially, the quality of explanations that a VLM can produce is shaped by its architecture. Among the most prominent medical VLMs developed for chest X-ray analysis are CXR-CLIP, which adapts contrastive language-image pre-training (CLIP), developed by OpenAI, to the radiology domain using datasets such as MIMIC-CXR, and BioViL and BioViL-T, which advance multimodal alignment through a radiology-specialised language model and temporally grounded reasoning across sequential imaging studies [12, 13, 14].

## 2.2 CXR-CLIP

CXR-CLIP extends the CLIP framework to the chest X-ray domain by introducing a prompt-based approach and fine-tuning on radiology-specific datasets including MIMIC-CXR [12]. It follows the same dual-encoder architecture as CLIP, where visual and textual modalities are processed by separate

encoders and projected into a shared 512-dimensional latent space for contrastive learning [15]. The image encoder employs a ResNet-50 to extract global visual features, while the text encoder uses BioClinicalBERT. CXR-CLIP's training combines the standard CLIP contrastive loss with image-level and text-level contrastive learning, enabling it to learn clinically relevant cross-modal representations.

### 2.3 BioViL / BioViL-T

BioViL was designed to address key limitations in earlier medical VLMs such as CXR-CLIP. It adopts a dual-encoder architecture that pairs CXR-BERT, a radiology-specialised text encoder pre-trained on large report corpora, with a ResNet-50 image encoder [13]. The image model is initialised through self-supervised pre-training on MIMIC-CXR, and both encoders project into a shared 120-dimensional latent space. Training combines global contrastive alignment (InfoNCE) with masked language modelling, promoting strong cross-modal alignment and interpretability. BioViL-T extends the framework into the temporal domain. It employs a hybrid CNN–Transformer multi-image encoder that integrates a ResNet-50 backbone with transformer-based temporal fusion [14]. By incorporating temporal embeddings, the model aligns corresponding anatomical regions across time, enabling it to capture disease progression.

### 2.4 Interpretability Techniques

Ensuring interpretability in medical VLMs is essential for clinical adoption, as their black-box nature undermines the transparent decision-making required in high-stakes settings. To address this, four complementary interpretability approaches are applied across both models. Attribution-based methods, including Grad-CAM, Grad-CAM++, and Integrated Gradients, generate saliency maps that highlight the image regions most influential to a model's prediction, providing a meaningful quantitative measure of localisation quality [16, 17]. Building on this, attention-based methods such as attention rollout and cross-modal alignment visualisation exploit the internal transformer architecture to examine how radiology-specific text tokens align with corresponding image patches across layers, producing spatially grounded explanations that are inherent to the model's structure [18].

Beyond spatial localisation, concept-based methods, specifically TCAV, evaluate the sensitivity of model predictions to clinically meaningful concepts such as cardiomegaly and pleural effusion by computing directional derivatives in the embedding space, yielding explanations that have demonstrated greater clinical plausibility than traditional heatmaps on MIMIC-CXR [19, 20]. Finally, counterfactual reasoning techniques affirm model robustness by generating targeted input modifications that alter predictions, thereby revealing causal relationships and identifying potential failure modes such as over-reliance on imaging artefacts or non-clinical visual cues [21, 22].

### 2.5 Aims & Objectives

**Core aims:**

1. Curate two high-quality subsets of the MIMIC-CXR-JPG dataset for single and temporal inference, removing duplicates, low-quality images and reports, incorrect view positions, and balancing pathology distributions.

2. Implement functional inference pipelines for BioViL-T and CXR-CLIP, ensuring both models operate as intended and obtain embeddings by running inference.

3. Implement the four interpretability families stated in Section 2.4: (1) attention-based, (2) attribution-based, (3) concept-based, and (4) counterfactual.

4. Ensure fair comparison between BioViL-T and CXR-CLIP using the single-inference pipeline, as well as between BioViL-T's single and temporal inference modalities.

**Stretch goals:**

1. Expose per-layer attention, per-patch projections, per-token text projections, and pre/post-fusion temporal representations from BioViL-T to enable architecture-specific methods (cross-modal alignment, four-quadrant temporal rollout) beyond the standard four-family taxonomy.
2. Develop a four-metric evaluation framework (LQ, SA, F, CP) with quantitative success criteria, enabling triangulation across method families rather than reliance on a single headline metric.
3. Diagnose scorer-geometry confounds in perturbation-based faithfulness testing, reporting blur and zero-mask modes side-by-side to separate methodological artefacts from genuine faithfulness signals.

# 3 Design

## 3.1 dataset curation strategy

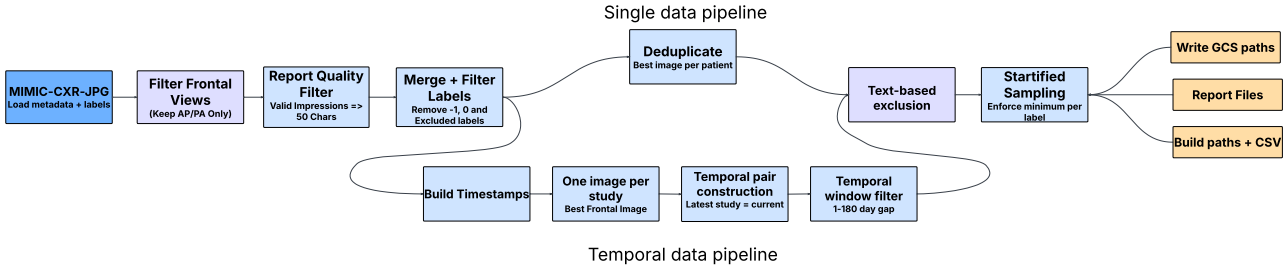



Figure 1: Design of the single and temporal datasets pipelines

### 3.1.1 Single-Inference Dataset Pipeline

The Single data pipeline shown in Figure 1 curates a subset of the MIMIC-CXR-JPG through a series of progressive filtering stages retaining only frontal-view CXRs, specifically posteroanterior (PA) and anteroposterior (AP), as these are the standard clinical views supported by both the models. Each remaining study is then assessed for report quality. Radiology reports are loaded to extract the findings and impression sections. Studies are retained only if they contain sufficient clinical details. CheXpert labels are then merged at the study level, providing pathology annotations for each study. The dataset only contains studies that are among the eight target conditions: pneumothorax, pleural effusion, support devices, consolidation, cardiomegaly, atelectasis, edema, and no finding. The eight target conditions were selected because they represent the most clinically prevalent and produce visually identifiable radiographic patterns making them well-suited for assessing the model's outputs. A deduplication step then reduces the dataset to one image per patient. This is done by using a scoring system that selects the highest-scoring image, ensuring that the most diagnostically clear radiograph is retained for each patient.

Finally stratified sampling is applied to enforce minimum representation across all eight target pathologies. This technique produces a clinically valid subset of the MIMIC-CXR mimicking the distribution of pathologies in the initial dataset. The subset of 10,000 was chosen, as a significantly larger subset would increase the computational cost.

### 3.1.2 Temporal Inference Dataset Pipeline

The temporal-inference pipeline shares the same initial filtering stages as the single-inference pipeline (Figure 1), ensuring consistent curation. It then adds temporal-specific steps: timestamps are con-

structed per study, and studies with missing timestamps are excluded to preserve valid temporal ordering. Deduplication is performed to one image per study (not per patient), since temporal inference requires two studies from the same patient; candidates are ranked by highest quality score, as in the single pipeline.

Temporal pairs are formed by selecting, for each patient, the most recent study as the current image and the closest prior study within a 1–180 day window, excluding same-day pairs due to minimal temporal information. Stratified sampling is applied with a different strategy from the single-inference pipeline to reflect the smaller pool of valid pairs, yielding a final subset of 2,000 temporal pairs. The smaller size reflects the stricter, more complex filtering process.

## 3.2 Inference Pipeline

### 3.2.1 BioViL-T

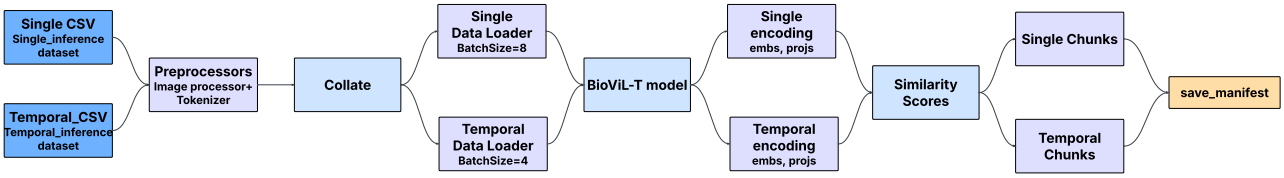



Figure 2: Design of the BioViL-T pipeline

The BioViL-T pipeline (Figure 2) transforms single and temporal subsets into structured embeddings supporting the interpretability tasks in Section 2.4. It provides separate single- and temporal-inference paths because the modalities differ: the single path processes one frontal CXR, while the temporal path processes a prior–current image pair via a custom data loader. Data are collated and preprocessed at collation time, allowing use of BioViL-T's native image processors and the CXR-BERT tokeniser from Hugging Face, ensuring transformations match pretraining [13].

In the single-inference path, embeddings and projections are extracted across all layers. The global embedding supports coarse image–text similarity, patch projections enable patch-level alignment with text tokens (required for cross-modal alignment and attribution), and attention maps support attention-based interpretability. The text encoder retains the global embedding, the full last hidden state for per-token projections, and attention maps and hidden states from every layer; per-token projections are necessary for token-level alignment visualisations. The temporal path extends this by encoding prior and current images and fusing them via BioViL-T's temporal module. It outputs four representations per sample: independent global embeddings, independent patch projections, current-image patch features, and temporally fused patch projections. Preserving all four enables comparison of pre- and post-fusion representations for interpretability. The pipeline also retains the temporal transformer's attention maps, the core mechanism for temporal reasoning. It computes three similarity scores per temporal sample (prior–text, current–text, prior–current) and one per single sample. Outputs are written as fixed-size chunks rather than a single file to prevent memory exhaustion during inference. These chunks are unified by a manifest file, which serves as the single entry point for downstream interpretability tasks.

### 3.2.2 CXR-CLIP

The CXR-CLIP inference pipeline (Figure 3) uses the same curated single-inference subset, partitioned at the subject level into a calibration split (80%) and a held-out test split (20%). Subject-level splitting prevents the same patient characteristics from appearing in both sets. The 80/20 ratio allocates more data to calibration because rare pathologies benefit from a larger pool of positive examples. Interpretability analysis operates on subsets of 10–200 cases per pathology and is not constrained by test-set size, so a larger calibration set strengthens decision thresholds. Preprocessing is attached to

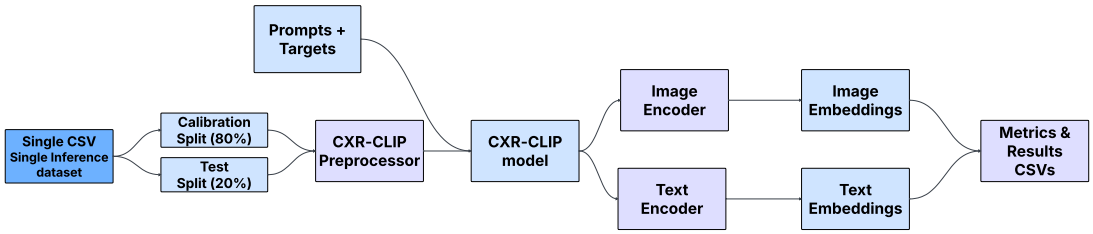



Figure 3: Design of the CXR-CLIP pipeline

the dataset rather than a collate function, concentrating tensor construction within the dataset's item accessor so each sample is returned as a preprocessed tensor. This is feasible because CXR-CLIP uses fixed prompt embeddings. After preprocessing, prompt ensembling encodes multiple phrasings and averages them to produce two composite directions per pathology, reducing sensitivity to wording and improving stability.

CXR-CLIP follows a dual-encoder design, where image and text encoders project into a shared latent space [12]. Image embeddings are extracted once per split and cached to disk since full forward passes are computationally expensive. Text embeddings are cached in memory and rebuilt from the fixed prompt dictionary. Both paths meet at similarity scoring, where each image embedding is compared with composite positive and negative directions per pathology using cosine similarity. For each image–pathology pair, the pipeline produces three values: mean positive similarity, mean negative similarity, and the margin, which avoids redundant similarity computations during analysis. As a zero-shot model, CXR-CLIP has no trained decision boundary, so margins are thresholded. Thresholds are fitted on the calibration split by maximising F1 on raw margins. F1 is used because it balances precision and recall under class imbalance, characteristic of the MIMIC-CXR dataset. This calibration–evaluation separation prevents conflating threshold fitting with model performance.

### 3.3 Interpretability Framework

Interpretability evaluation of vision–language models requires multiple complementary methods. Attention-based methods analyse internal attention weights to identify influential spatial regions and text tokens. For BioViL-T, attention rollout propagates self-attention across transformer layers to produce patch-level heatmaps for single-image inference, highlighting spatial sensitivity [18]. For temporal inference, rollout is decomposed into four quadrants (current→current, current→previous, previous→current, previous→previous) to assess whether temporal modules capture inter-study change. Cross-modal alignment is also applied to BioViL-T, leveraging per-patch projections for token-level queries. As CXR-CLIP's ResNet-50 encoder does not expose self-attention, a similarity-relevance mapping is used [13, 12]. Attribution-based methods trace outputs back to input pixels to produce saliency maps. Grad-CAM and Grad-CAM++ are applied to both models at matched architectural depth, enabling direct comparison; Grad-CAM++ is included for its sharper localisation. Using both ensures observed differences reflect model behaviour rather than method choice. Integrated Gradients serves as a theoretically grounded baseline, providing pixel-level attributions [16, 17].

Concept-based methods assess whether latent spaces encode clinical concepts. TCAV learns linear directions in embedding space that separate positive and negative concept samples, measuring alignment of model representations. It is applied to both models on the same pathology set to ensure fair comparison [19]. Prototype retrieval complements TCAV by identifying nearest neighbours in embedding space, revealing clinically meaningful clustering at the instance level. Cross-modal retrieval (BioViL-T only) further evaluates whether its shared latent space aligns images and text semantically [20]. Counterfactual methods evaluate prediction sensitivity to controlled input changes [21]. Textual negation tests cross-modal grounding by replacing clinical text with its negation: BioViL-T uses report substitution with re-evaluated similarity, while CXR-CLIP swaps positive and negative prompts; both should show decreased similarity. Generic prompt substitution replaces disease-specific

prompts with neutral ones to assess disease understanding, as BioViL-T does not rely on prompt sets. Temporal counterfactuals compare correct versus mismatched priors to test temporal module utility. Latent-space counterfactuals perturb embeddings along concept activation vectors to measure causal effects. Attention suppression removes attention pathways, combined with rollout, to test their necessity for spatial and temporal reasoning [22].

#### 3.3.1 Evaluation Metric & Success Criterion

Interpretability outputs carry incompatible units, so no single metric captures both spatial fidelity and concept-level meaning. Four complementary families are applied, each with a quantitative success criterion grounded in the property it measures: where the model looks, what it represents, whether those regions drive predictions, and whether the explanations are clinically defensible.

Localisation Quality (LQ) measures pointing-game hit-rate against CheXmask anatomical proxy masks, with success requiring per-pathology rates that exceed a uniform random-peak baseline on the same mask, ensuring hits reflect model-driven localisation rather than proxy coverage. Semantic Alignment (SA) combines heatmap-shape descriptors (entropy, peak-to-mean ratio, sparsity), reported as distributions for comparative interpretation, with TCAV concept-sensitivity scores; success requires TCAV above 0.5, the chance level under random concept directions, with scores above 0.7 treated as strong concept encoding. Faithfulness (F) reports direction-of-effect criteria rather than absolute AUCs, because BioViL-T's cosine-similarity scorer and CXR-CLIP's prompt-margin scorer differ by an order of magnitude in scale. Success requires insertion AUC $>$ deletion AUC, saliency-guided drops exceeding ten matched random-mask drops in either zero or blur mode, and per-sample win-rates above 50%; Integrated Gradients serves as a third-party baseline so effects can be attributed to model behaviour rather than attribution method. Clinical Plausibility (CP) applies four artefact heuristics: border focus, rib overfocus, centre bias, and activation spread; with centre bias excluded as a saturated check, success on the remaining three requires a pass rate $\geq 90\%$.

## 4 Implementation

### 4.1 Dataset curation

Both subsets are curated from the **MIMIC-CXR-JPG v2.1.0** on PhysioNet using the metadata table, CheXpert label table, and free-text radiology reports [23].

#### 4.1.1 Single-inference pipeline

The pipeline begins by loading metadata and CheXpert tables into a pandas DataFrame. From 377,110 rows, a boolean mask on **ViewPosition** restricts the dataset to 243,334 frontal (PA or AP) studies, and duplicate **dicom_id** rows are removed. Radiology reports are attached via a **load_report()** helper, returning None if missing; such rows are dropped. Reports under 100 characters are removed, then the FINDINGS and IMPRESSION sections are extracted into the dataframe. A validation step excludes impressions that are empty, contain only underscores, or are under 50 characters due to low content. CheXpert labels are joined to the meta data frame. Rows with uncertain (-1) or excluded labels are discarded. Additionally, rows where any of five excluded labels (Enlarged Cardiomediastinum, Fracture, Lung Lesion, Pleural Other, Pneumonia) are positive (1) are removed as such samples would cause ambiguity and disrupt interpretability analysis. Rows with no positive target labels are discarded, while the "No Finding" label serves as the control group. Finally, a text-based exclusion compares impressions against a dictionary of excluded labels.

The current dataframe is deduplicated using **deduplicate_mimic_cxr()**. A quality score is computed from pixel area with a penalty from **PixelSpacing**, and a view rank prefers PA over AP due to clinical preference. Acquisition datetime is reconstructed from **StudyDate** and the integer portion of **StudyTime** for tiebreaking. The frame is sorted within each **(subject_id, study_id)** by view rank,

quality score, and recency, retaining the best row per study reducing the set to 23,340 rows. Stratified sampling then selects 10,000 patients, over-representing two rare pathologies (Pneumothorax and Consolidation, 900 each) relative to the remaining six more prevalent ones (1,200 each). The remainder is filled via uniform sampling to reach 10,000 rows, saved to **Dataset/Single/single_dataset.csv** retaining the columns: **subject_id**, **study_id**, **dicom_id**, CheXpert labels, **image_path**, and **report_path**. Two export artefacts complete the pipeline: GCS image URIs are written to **image_paths.txt** and used to download CXR JPGs into **Dataset/Single/images/**, preserving structure; matching report .txt files are copied into **Dataset/Single/Reports/**. A final pass constructs absolute image and report path columns.

#### 4.1.2 Temporal-inference pipeline

The temporal notebook reuses the initial logic from the single pipeline to ensure consistency. Differences lie in acquisition time handling, deduplication, pair construction, and sampling stratification. An acquisition time column is first created by concatenating **StudyDate** with the integer portion of **StudyTime**, zero-padded to six digits and parsed via a format string. Rows failing parsing are dropped, as temporal pairing requires reliable ordering. The function **deduplicate_to_one_image_per_study()** reduces to one image per study using similar quality and view-rank heuristics as the single pipeline, but at the study level, retaining multiple studies per patient. Pair construction filters the frame and sorts by **subject_id**, acquisition time, and **study_id**, selecting the latest eligible study per patient as the current. Each current row is joined with prior rows from the same patient with earlier acquisition times. Gaps are computed within $1 \leq \text{days} \leq 180$ to avoid same-day repeats, and the closest prior is selected.

The excluded-text scan is applied only to current impressions, as priors serve as visual context; filtering them would remove valid pairs. This reduces pairs from 8,252 to 7,099. Stratified sampling follows a priority scheme due to the smaller, skewed pool. All rows labelled Consolidation or Pneumothorax are retained, while the remaining 2,000-pair budget is proportionally distributed across other strata, flooring rare pathologies while preserving prevalence. The result is saved as **/Dataset/Temporal/temporal-dataset.csv**. The sampled frame is then cleaned by retaining only **subject_id**, **prior_study_id**, **prior_dicom_id**, **current_study_id**, **current_dicom_id**, **gap_days**, **current_report_text**, CheXpert labels, and three path columns: **prior_image_path**, **current_image_path**, and **report_path**. Two export artefacts complete the pipeline: GCS image URIs are written to image paths **image_paths_temporal.txt** and used to download CXR JPGs into **Dataset/Temporal/images**, preserving structure; matching report .txt files for current studies are copied into **Dataset/Temporal/Reports/**.

### 4.2 The inference pipeline

#### 4.2.1 BioViL-T pipeline

The BioViL-T implementation begins by cloning all the image-encoder source files from Microsoft's official hi-ml repository (hi-ml-multimodal/src/health_multimodal/image/model) into **models/biovil_t**, as the wrapper class in **models/load.py** depends on the image encoder constructor in pretrained.py and the **VisionTransformerPooler** class in transformer.py [24].

Dataset access is provided by two classes in **data_loader.py**. **SingleInferenceDataset** class validates the required image paths and columns. **TemporalDataset** class follows the same structure but exposes **prior_image_path**, **current_image_path**, **current_report_text**, and **gap_days**. Both then return raw paths and strings, deferring tensor construction to the collate stage. Preprocessors in **preprocessors/biovil_t.py** are loaded through **load_biovil_preprocessor()**: the image mode returns **create_chest_xray_transform_for_inference()**, which resizes the CXRs to 512 and applies a centre crop to 448, matching the pre-training transform; the text mode returns the Hugging Face AutoTokenizer for BioViL-T (CXR-BERT). DataLoaders use two collate functions in **collate/biovil_t.py**, **collate_biovil_single()** and **collate_biovil_temporal()** so that the image processor and tokenizer

are bound once at construction. Images are loaded through hi-ml's load_image utility, transformed, and stacked. The temporal collate stacks prior and current images into separate tensors and tokenises the current-study report only. Both assemble a meta dictionary of identifiers and paths and a labels dictionary of float tensors.

The wrapper in **models/biovil_t/load.py**, **BioVILTWrapper** class, composes two sub-models. The text model is instantiated through AutoModel.from_pretrained, exposing cls projection head for token-level projection. The image model is instantiated through **get_biovil_t_image_encoder()**, returning ResNet-50 coupled with the temporal **VisionTransformerPooler**. Because the pooler is nested inside the image model, the wrapper locates it by iterating image model module and retaining the first pooler instance.

The main pipeline file **pipelines/biovil_t_pipeline.py** containing the inference logic consists of **extract_single_outputs_chunked()** that runs under torch.inference_mode() and calls **encode_sin -gle_image_intermediates()** per batch to obtain the global embedding, patch projections, and per-block attention maps. **encode_text_intermediates()** yields the CLS projected embedding for image–text similarity and the last hidden state, which is passed to **project_text_hidden_states()** for per-token projections. **extract_temporal_outputs_chunked()** encodes prior and current images independently, then calls **encode_temporal_intermediates()**. Temporal intermediates are captured via a forward pre-hook that intercepts the pooler's arguments; the hook is removed and the pooler is re-invoked with **return_attentions=True**. Four image representations are preserved per sample (independent prior and current globals, independent patch projections, current-image patch features, and fused patch projections), alongside temporal attentions and three similarity scores: prior–text, current–text, prior–current.

Both routines flush at target sizes (100 single, 50 temporal) via **save_single_chunk()** or **save_tem -poral_chunk**, stacking uniform tensors and keeping lists for variable-length fields. **save_manifest()** records chunk layout, and cache-aware wrappers **load_or_extract_biovil_t_single_embeddings()** and **load_or_extract_biovil_t_temporal_embeddings()** skip extraction when the manifest exists. **load_single_samples()**, **load_temporal_samples()** reconstruct flat sample lists and partition them into positive and negative pools under a seeded sampler for reproducibility.

### 4.2.2 CXR-CLIP pipeline

The CXR-CLIP implementation begins by cloning the official repository into **models/cxr_clip**; the MIMIC-CXR (ResNet-50) checkpoint listed in readme.md is also downloaded, as it stores the configurations required for both model construction and image preprocessing. The wrapper in **models/cxr_clip/load.py** prepends the repository path to sys.path through **_add_repo_to_path**, allowing official modules to be imported without modification [12].

Preprocessing is defined in **preprocessor/cxr_clip.py** via **load_cxr_clip_preprocessor()**, which reads the checkpoint file stored in the locally and builds the preprocessor using **load_transform()** and **transform_image()**, ensuring inference preprocessing matches training. The returned callable converts a PIL image to RGB, applies the transform, and outputs a [3, H, W] tensor. Prompts are built for each pathology in the dataset in a format similar to the ones during training for a fair evaluation. The inference dataset, **CXRClipInferenceDataset** class in **data_loader.py**, fills missing labels with zero and enforces label consistency by setting No Finding to zero when any abnormal label is positive, applied in the constructor to prevent contradictory labels from biasing the threshold fit. **models/load_cxr_clip** loads the checkpoint to the repository's build_model and load_tokenizer utilities. Tokenizer and text-encoder cache paths are overridden with a locally resolved Hugging Face cache so downstream calls resolve to a writable location. Three helpers are exposed: **encode_image()** passes a batched tensor through the image projection head and L2-normalises; **encode_text()** tokenises the prompts and L2-normalises; **score_prompts()** returns cosine similarity between an image and a list of prompts.

The pipeline in **pipelines/cxr_clip_pipeline.py** threads a single dictionary from **init_cxr_cli -p_pipeline()** through all functions. Initialisation builds a text embedding cache via **build_te-**

**xt_embedding_cache** from **utils/prompts.py**, amortising the text-encoder forward pass. **score_image_embedding()** L2-normalises an image embedding and records the margin (mean positive minus mean negative similarity) to avoid recomputation during calibration. **extract_dataset_image_embeddings()** extracts image embeddings once per split into a NumPy array with metadata; **get_or_create_dataset_image_embeddings()** saves and reloads these, avoiding repeated ResNet-50 passes. Text embeddings remain in memory as the prompt set is fixed.

Because CXR-CLIP is zero-shot, margins must be thresholded. In **calibration/calibration.py**, **find_best_threshold()** sweeps 400 thresholds per pathology, selecting the F1-maximising value to balance precision and recall under imbalance. **fit_thresholds_from_results_df()** applies this to the seven abnormal pathologies; **add_calibrated_predictions_to_df()** appends per-pathology predictions, and No Finding is derived as the negation of any abnormal prediction rather than thresholded, maintaining label consistency. **evaluate_predictions_from_df()** computes precision, recall, specificity, accuracy, balanced accuracy, and F1. **run_cxr_clip_calibration_and_evaluation()** partitions subject identifiers into calibration (80%) and test (20%) splits under a fixed random state; thresholds fit on calibration alone are applied unchanged to the test split, without which performance would conflate discriminative quality. Raw and calibrated results, threshold summaries, and per-pathology metrics are written as CSVs.

## 4.3 Interpretability Techniques

### 4.3.1 BioViL-T

The interpretability techniques for BioViL-T are implemented in the **interpretability/biovil_t** directory and comprise attention-based, gradient-based, concept-based, and counterfactual explanations.

The attention-rollout module **attention_rollout.py** propagates self-attention across the layers of the **VisionTransformerPooler** exposed by the image encoder. **fuse_heads()** reduces each per-layer tensor from [B, H, N, N] to [B, N, N] by averaging across heads. **normalize_attention()** then adds an identity matrix and row-normalises, accounting for the residual connections that bypass the attention block. **attention_rollout** chains the per layer matrices, yielding a single [B, N, N] matrix that approximates how much information tokens contribute, once all attention layers are traversed. Two spatial helpers adapt the rollout matrix to BioViL-T's patch grid. **spatial_rollout_no_cls()** collapses the matrix along its first axis and reshapes the resulting vector to a 14x14 grid reflecting BioViL-T's pooler that has no CLS token. **upsample_heatmap()** then applies bilinear interpolation to the native 448x448 input resolution so that the heatmap can be overlaid in the CXR. For temporal inference, **split_temporal_rollout()** partitions the rollout matrix into four quadrants corresponding to the directions of cross-study attention. Each quadrant is then reduced to a 14x14 heatmap via **row_to_patch_heatmap()**. This is to make sure that the temporal module is attending across studies.

**cross_modal_alignment.py** computes patch-token similarity from the per-patch and per-token projections. **_compute_alignment_map()** L2-normalises both the projections, computes the similarity cube, and returns it alongside the valid token list. Three query primitives are built on top of this cube: **token_heatmap()** returns the 14x14 map for a single token; **phrase_heatmap()** extends this to a multi-token phrases by locating phrase sequence with **find_subsequence()**; and **topk_tokens_for_patch_and_token()** inverts the query direction, answering which token strongly aligns with the given patch location.

The gradient-based attribution module **grad_cam.py** wraps forward and full-backward hooks around a chosen convolutional layer of the image encoder. **find_target_layer()** resolves the target layer by scanning the module tree for the last conv2d layer. The **BioViLTGradCAM** class captures activations on the forward pass and gradient tensors on the backward pass, reproducing the standard Grad-CAM. Per-channel weights are computed and applied to the activations to produce a weighted sum, passed through a ReLU, and min-max normalised per sample. The target scalar is the image-text similarity computed as the dot product of the L2-normalised global image and text embedding.

**BioViLTGradCAMPlusPlus** extends this by retaining activations so that their gradient can be queried. A guarded division handles zero-gradient regions, and the final weights aggregate alpha-scaled positive gradients. Both classes share the same hook lifecycle and the same normalisation scheme. Both are wrapped by **init_grad_cam()** and **init_grad_cam_pp()**.

TCAV is implemented in **tcav.py**. **train_linear_cav()** fits a logistic regression with Adam and weight decay on concatenated positive and negative embeddings and extracts the weight vector, and L2-normalises it to obtain a CAV. **tcav_score_from_embeddings()** reports the fraction of samples whose embeddings project positively onto the CAV, and **similarity_tcav_score()** estimates a finite-difference directional derivative by perturbing each embedding along the CAV and measuring whether the image-text similarity increases or decreases. For **prototype_retrieval.py**, **retrieve_prototypes()** L2-normalises a query image embedding, scores it against every sample's image embeddings via a dot product, and returns the top-k sorted by similarity.

**counterfactuals.py** implements four counterfactual families. The first family is textual counterfactuals, which substitute a report with a modified version. **evaluate_textual_counterfactual()** tokenises both the original and counterfactual reports, computes the similarity for each, and returns the score drop. Temporal counterfactuals, implemented in **evaluate_temporal_counterfactual()**, compare the true temporal score against a mismatched prior score and a prior-less current to assess the temporal module. Latent counterfactuals perturb the image embedding along a CAV and measure the resulting similarity delta. **latent_counterfactual_edit()** computes $e + \epsilon c$ to nudge the embeddings towards the concept and $e - \epsilon c$ to nudge the embeddings away from the concept and re-normalises, and **evaluate_latent_counterfactual_similarity()** reports the similarities before and after edits. Attention suppression is maintained in **attention_control.py** through **get_active_attention_suppressor()** and a **use_attention_suppressor()**, which swap the active suppressor callable in and out of a module-level variable. Three suppression targets are supported through **AttentionSuppressionConfig**: the topk target, the heads target which ablates specific attention heads, and the temporal-quadrant target which ablates one of the four quadrants of the temporal attention matrix.

### 4.3.2 CXR-CLIP

The **interpretability/cxr_clip** directory implements similarity-based relevance mapping, Grad-CAM and multiple counterfactual strategies.

Similarity relevance map **similarity_relevance.py** uses **_get_image_backbone_and_projection()** which accesses the model's backbone and image projection head from the repository wrapper. **get_last_conv_features()** forwards the image through layer 4 of the ResNet, yielding a feature map. **generate_similarity_relevance_map()** then scores every cell against the mean positive and mean negative prompt embeddings, computes the margin, reshapes the result, and bilinearly upsamples to the input resolution. This produces a pathology-specific relevance heatmap.

Grad-CAM and Grad-CAM++ in **grad_cam.py** follow the same forward/backward hook pattern as BioViL-T. The target scalar is built from prompt embeddings using **_get_label_text_embeddings()** retrieves the mean positive and negative embeddings. **_forward_to_spatial_features()** runs the image through the ResNet up to Layer4, and **_target_score_from_spatial()** converts the spatial features into a global image embedding through the projection head, scores it against the positive and negative directions. **generate_gradcam()** and **generate_gradcam_plus_plus()** are used to generate gradcam heatmaps for standard and the plus plus variant.

**counterfactuals.py** implements four counterfactual families for CXR-CLIP. For negation counterfactuals, **_evaluate_negation_counterfactual()** swaps the positive and negative prompt lists for a pathology and recomputes the margins. For generic-prompt counterfactuals, **evaluate_generic_prompt_counterfactual()** replaces the disease-specific prompts with two semantically neutral templates and measures how the margin survives the substitution. Both are thin wrappers around **evaluate_prompt_counterfactual()**. For latent counterfactuals, **cxr_clip_latent_counterfactual()** normalises the image embedding, perturbs it along the CAV, re-normalises, and reports the margin under each edit. Because CXR-CLIP evaluates against a fixed prompt set, the perturbed embedding is

scored through **score_image_embedding()** using the same cached positive and negative prompt embeddings as the main pipeline. A masking counterfactual **evaluate_masking_counterfactual()** is also included as a faithfulness probe: given a heatmap, it identifies the top-k% most relevant pixels, constructs a mask at that threshold, and recomputes the margin.

### 4.4 Evaluation Framework

The **evaluation/** directory contains Localisation Quality, Semantic Alignment, Faithfulness, and Clinical Plausibility evaluation implementations.

The **localisation_quality.py** module implements the Pointing Game against CheXmask anatomical segmentations, chosen as ground truth for the MIMIC-CXR-JPG dataset. **load_chexmask_mimic_csv()** streams the CheXmask CSV in 50,000-row chunks retaining only rows whose **dicom_id** appears in the curated subsets; this was necessary due to high memory pressure. **decode_rle()** converts each RLE string into a binary tensor, **build_chexmask_lookup()** assembles a per DICOM dictionary, and **attach_chexmask_to_samples()** joins the lookup onto the sample lists by DICOM identifier. Because CheXmask annotates anatomy, **build_proxy_mask()** maps each target pathology to an anatomically plausible region. The pointing game itself is implemented in **pointing_game_mask()**: **ensure_same_size()** bilinearly resizes the heatmap to the mask resolution and normalises to [0,1], **get_peak_point()** returns the argmax, and a hit is recorded if the peak falls inside the proxy mask.

**semantic_alignment.py** quantifies the heatmap quality and concept sensitivity by rewarding a correctly placed peak. **heatmap_entropy()** treats it as a probability distribution and computes Shannon entropy; **peak_localisation_score()** returns the max-to-mean ratio, capturing how strongly the peak stands out; and **heatmap_sparsity()** reports the fraction of pixels above 0.7. The heatmap quality is then evaluated using the evaluation function, similarity-relevance maps for CXR-CLIP and cross-modal alignment for BioViL-T. Concept sensitivity is handled by **build_tcav_concepts()**, which for each pathology calls **select_sample_by_label()** to partition into positives and negatives, **build_cav_from_samples()** to fit a linear CAV, **tcav_score_from_embeddings()** to compute the projection score and, when text embeddings exist, **similarity_tcav_score()** for the direction derivative on image-text similarity.

The **faithfulness.py** module is built around a scorer abstraction **make_biovilt_scorer()**; it tokenises a fixed report once and scores an image tensor against it, while **make_cxr_clip_scorer()** scores against the calibrated prompt margins. **evaluate_deletion_insertion()** sweeps a fraction from 0.0 to 1.0 across 11 steps; at each step **top_k_fraction_mask()** identifies the top-k% relevant pixels, a deletion image zeroes them, and an insertion image begins from a zero or blurred baseline. **evaluate_masking_vs_random()** complements this with a single-fraction comparison against 10 random masks at fraction 0.2, exposed in both zero and blur modes as zeroing introduces sharp edges and blurring preserves low-frequency structure. Integrated Gradients is included as a baseline, **_biovilt_ig_forward()** and **_cxr_clip_ig_forward()**, return the scalar that each model is trained to optimise. **compute_integrated_gradients** invokes Captum's `integrated_gradients`. **evaluate_faithf-ulness_for_heatmap()** composes deletion-insertion, zero-mode masking, and blur-mode masking into a single call.

**clinical_plausibility.py** catches heatmaps that are clinically implausible because they concentrate on borders, fixates on rib stripes, or collapses to a single pixel of a CXR. Four heuristics implement this scrutiny. **check_border_focus()** constructs a 10% border mask and flags heatmaps where border activation exceeds 50% of the total, catching fixation on text overlays, date stamps, or collimation edges. **check_rib_overfocus()** sums activation per row and flags cases where the top 5% of rows contain more than 40% of the activation, catching fixation on horizontal rib edges. **check_centre_bias()** flags both excessively centred heatmaps and excessively de-centred ones. **check_activation_spread()** flags values below 0.01 or above 0.5. **evaluate_artefact_checks_batch()** invokes all four on a single heatmap across a sample list.

### 4.5 Experimental Setup

Experiments are run in **run_experiments.ipynb**. Setup proceeds by first creating a directory named **interpretable-cxr-models** following the structure in Section 4; ensuring dataset curation has produced valid **single_dataset.csv** and **temporal_dataset.csv** with correct image and report paths, and feeding these CSVs into downstream functions; performing imports; extracting BioViL-T embeddings via **load_or_extract_biovil_t_single_embeddings()** and **load_or_extract_biovil_t_temporal_embeddings()**, with embeddings cached as chunks under **embeddings/biovil_t** and reconstructed into flat sample lists; initialising CXR-CLIP via **init_cxr_clip_pipeline()** and running **run_cxr_clip_calibration_and_evaluation()** with **test_size = 0.2**, which returns calibration thresholds, the test split, and the **cxr_clip_samples** list; and attaching CheXmask, where **MIMIC-CXR-JPG.csv** is streamed through **load_chexmask_mimic_csv()** with **min_dice_mean = 0.7** and joined onto the sample lists to produce matched single and temporal samples.

Three comparisons are run: BioViL-T (single) vs. CXR-CLIP (single), BioViL-T (single) vs. BioViL-T (temporal), and attribution methods within each model. Localisation Quality (LQ) and Semantic Alignment (SA) are pathology-anchored, as both require a clinical concept, whereas Faithfulness (F) and Clinical Plausibility (CP) are model-level diagnostics reported once per model with method-level breakdowns. Single-image experiments draw from the 10,000-sample subset (Section 4.1.1) and temporal experiments from the 2,000-pair subset (Section 4.1.2). CXR-CLIP is evaluated on the held-out 20% test split (Section 4.2.2), while BioViL-T uses the full single-inference set. The Pointing Game is restricted to DICOMs with CheXmask, and LQ and SA are reported only for Cardiomegaly, Pleural Effusion, and Pneumothorax. Sample allocations are fixed to keep comparisons fair across pipelines. LQ uses **balance_pathology_sample()** to set $n = \min(\text{single_pos}, \text{temporal_pos})$ per pathology, so BioViL-T (single), BioViL-T (temporal), and CXR-CLIP are evaluated on identical pools; Grad-CAM and Grad-CAM++ run on the matching single-pathology cases drawn from **test_results_cal**. SA uses 100 positive and 100 negative samples per pool.

## 5 Results and Evaluation

### 5.1 Localisation Quality [LQ]

Localisation quality is measured against CheXmask anatomical proxy masks: a heatmap "hits" if its peak pixel falls inside the anatomically plausible region for the pathology in question. Results across the six attribution methods are summarised in **Table 1**.

Table 1: Pointing Game results across attribution methods on Cardiomegaly, Pleural Effusion, and Pneumothorax.

| Method | Pool | Overall | Cardiomegaly | Pleural Effusion | Pneumothorax |
|---|---|---|---|---|---|
| BioViL-T Spatial Rollout | 885 | 0.1299 | 49/199 | 33/539 | **33/147** |
| BioViL-T Temporal Rollout | 885 | 0.1017 | 20/199 | **52/539** | 18/147 |
| BioViL-T Grad-CAM | 694 | 0.1974 | 102/284 | 20/341 | 15/69 |
| BioViL-T Grad-CAM++ | 694 | 0.1657 | 88/284 | 11/341 | 16/69 |
| CXR-CLIP Grad-CAM | 694 | 0.2622 | 169/284 | 4/341 | 9/69 |
| CXR-CLIP Grad-CAM++ | 694 | **0.3501** | **231/284** | 7/341 | 5/69 |

CXR-CLIP Grad-CAM++ achieves the highest overall pointing score (**0.3501**) and outperforms every BioViL-T method, primarily through dominant Cardiomegaly performance (**81.3%**). BioViL-T Grad-CAM and Grad-CAM++ score **0.1974** and **0.1657**, while spatial and temporal rollouts trail at **0.1299** and **0.1017**. Within each model, gradient-based methods outperform attention-based ones at matched architectural depth, suggesting gradients carry stronger spatial signal than self-attention propagation. The Grad-CAM++ extension benefits CXR-CLIP **(0.2622 → 0.3501)** but degrades BioViL-T **(0.1974 → 0.1657)**, indicating that alpha-weighted positive-gradient aggregation sharpens

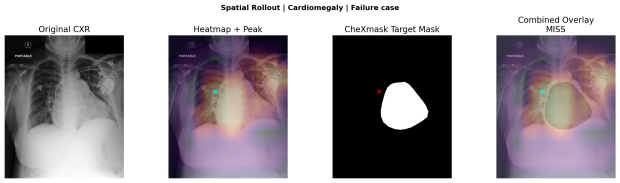



Figure 4: Spatial rollout failure case (Cardiomegaly).

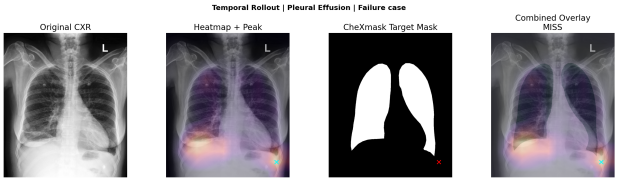



Figure 5: Temporal rollout failure case (Pleural Effusion).

ResNet-50 saliency while fragmenting BioViL-T's already-concentrated transformer attributions.

The overall scores hide stark per-pathology asymmetry. Both CXR-CLIP methods localise Cardiomegaly extremely well **(59.4%, 81.3%)** but collapse on Pleural Effusion **(1.2%, 2.1%)** and Pneumothorax **(13.0%, 7.2%)**. This is consistent with Cardiomegaly being a large central structure whose proxy mask covers a substantial fraction of the central image, so any centred heatmap registers as a hit, while peripheral pathologies penalise central concentration. BioViL-T's attention rollouts show an inverted profile and outperform CXR-CLIP on peripheral pathologies. Spatial rollout reaches **22.4%** on Pneumothorax **(vs 7.2%)** and temporal rollout **9.6%** on Pleural Effusion **(vs 2.1%)**. The Pleural Effusion improvement under temporal rollout is the only case where the temporal module outperforms its single-image counterpart, suggesting change-over-time signals contribute meaningful localisation, though the temporal module underperforms on Cardiomegaly and Pneumothorax. Qualitatively, rollout misses on Cardiomegaly diffuse across both lung fields rather than localising centrally, while Pleural Effusion misses cluster centrally, indicating centre bias.

#### 5.1.1 Synthesis

The two model families exhibit complementary behaviour. CXR-CLIP's gradient methods excel on large central pathologies where diffuse heatmaps benefit from anatomical overlap with the image centre, while BioViL-T's attention rollouts distribute hits more evenly across peripheral regions where central concentration is penalised. CXR-CLIP's headline score therefore partly reflects centre-bias rather than disease-specific localisation, a finding later evidenced by the centre_bias failure in Section 5.4. BioViL-T's lower headline scores conceal more anatomically discriminating behaviour.

### 5.2 Semantic Alignment [SA]

#### 5.2.1 Heatmap quality

Three scalar descriptors in **Figure 6** are computed on each normalised heatmap: entropy (lower = more concentrated), peak score (higher = stronger peak relative to mean activation), and sparsity (lower = a smaller high-activation region).

The most striking finding is the entropy gap: BioViL-T's median entropy is **5.19**, less than half of CXR-CLIP's **10.69**, and the two distributions do not overlap at all. This is not a marginal effect. BioViL-T's cross-modal heatmaps concentrate the bulk of their activation on a small fraction of patches, whereas CXR-CLIP's similarity relevance maps spread activation across a much wider region of the image. BioViL-T's entropy is tightly distributed around its median, while CXR-CLIP's distribution is similarly tight but at a fundamentally higher level, indicating the difference reflects systematic architectural behaviour rather than per-sample variance. The peak score and sparsity show much smaller cross-modal difference. CXR-CLIP edges out BioViL-T on peak score **(median 2.17 vs 2.04)**, meaning that despite its higher overall entropy, CXR-CLIP's single brightest pixel still stands out marginally more sharply against its own mean activation. Sparsity tells a similar story **(BioViL-T 0.15, CXR-CLIP 0.18)** with BioViL-T producing slightly more selective high-activation regions.

These three results together describe a specific shape difference. BioViL-T produces heatmaps with a small concentrated focal region surrounded by near-zero activation everywhere else (low entropy, low

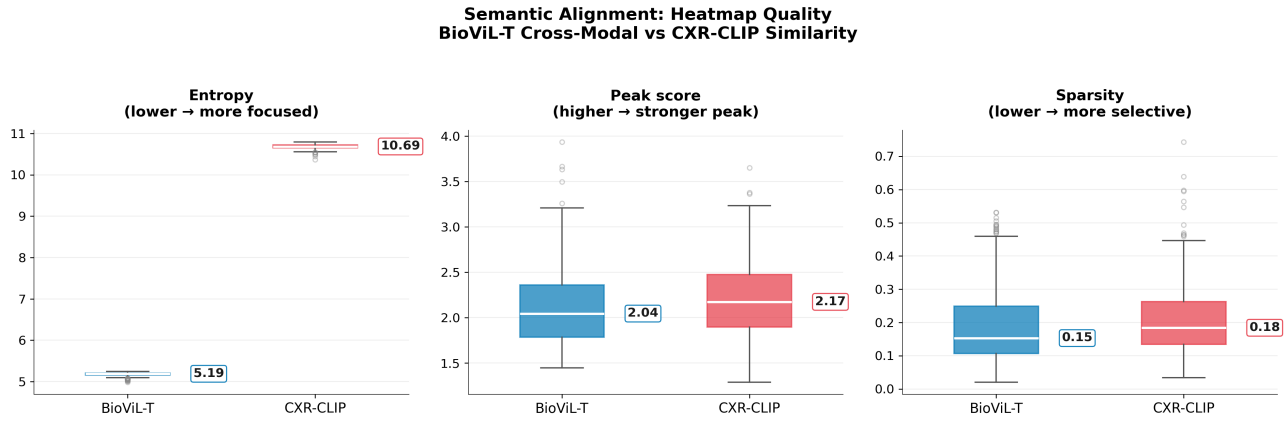



Figure 6: Heatmap quality Box plot

sparsity, moderate peak). CXR-CLIP produces heatmaps with a larger active region that is relatively flat but with a slightly stronger single peak (high entropy, slightly higher sparsity, slightly higher peak). These results are consistent with how BioViL-T's cross-modal alignment maps are computed by each patch token against the report embeddings in a shared latent space. CXR-CLIP's similarity-relevance maps are computed against a prompt-ensemble margin derived from a ResNet-50 backbone, where the global pooling structure spreads similarity across spatially adjacent locations.

### 5.2.2 TCAV

Concept activation vectors were trained on 100 positives and 100 negatives per model per pathology, with models scored on the resulting CAVs. Table 2 reports the embedding TCAV and, where available, the similarity TCAV only for BioViL-T because CXR-CLIP does not persist per-sample text embeddings.

Table 2: TCAV scores per pathology and model: embedding TCAV for both models; similarity TCAV (BioViL-T only).

| Pathology | Model | N_pos | N_neg | Embedding_TCAV | Similarity_TCAV |
|---|---|---|---|---|---|
| Cardiomegaly | BioViL-T | 100 | 100 | 0.5750 | 0.7100 |
| Cardiomegaly | CXR-CLIP | 100 | 100 | 0.6100 | N/A |
| Pleural Effusion | BioViL-T | 100 | 100 | 0.4800 | 0.6650 |
| Pleural Effusion | CXR-CLIP | 100 | 100 | 0.4550 | N/A |
| Pneumothorax | BioViL-T | 100 | 100 | 0.7350 | 0.4700 |
| Pneumothorax | CXR-CLIP | 100 | 100 | 0.5900 | N/A |

Embedding TCAV reveals a more nuanced picture than heatmap quality. BioViL-T outperforms CXR-CLIP on Pneumothorax **(0.7350 vs 0.5900)** by a substantial margin, while it underperforms on Cardiomegaly **(0.5750 vs 0.6100)**. On Pleural Effusion the two models score within 0.025 of each other **(BioViL-T 0.4800, CXR-CLIP 0.4550)**. BioViL-T's Cardiomegaly score of **0.5750** is at the random baseline suggesting that the Cardiomegaly direction in BioViL-T's image embedding space does not separate positive from negative cases beyond what would be expected, given that Cardiomegaly is anatomically the most distinctive of the three pathologies, which makes this finding particularly surprising. This pattern mirrors the results in Section 5.2: CXR-CLIP localises Cardiomegaly best in the Pointing Game, and CXR-CLIP also encodes Cardiomegaly more strongly in TCAV. But the convergence breaks down for the other two pathologies. Conversely, BioViL-T localises Pneumothorax under spatial rollout at 22% and encodes it strongly in TCAV **(0.735)**, suggesting the model has internalised the concept of Pneumothorax in its latent space without translating that concept into a spatial heatmap.

The similarity TCAV column, only for BioViL-T, gives further evidence that BioViL-T's image and text embeddings are consistent: **0.7100** for Cardiomegaly, **0.6650** for Pleural Effusion and **0.48** for Pneumothorax. These numbers diverge from the corresponding embedding TCAV scores in pathology-dependent ways, indicating that the alignment between visual concept representation and image-text similarity is itself pathology-dependent. The Pneumothorax similarity score of **0.4700** is notable, suggesting that BioViL-T's text encoder may not reciprocate the Pneumothorax direction as strongly as its image encoder does.

#### 5.2.3 Synthesis

Heatmap quality is dominated by the entropy gap **(BioViL-T 5.19 vs CXR-CLIP 10.69)** with non-overlapping distributions, confirming that BioViL-T's cross-modal alignment maps concentrate activation on small focal region while CXR-CLIP's similarity-relevance maps spread it across a wider area. Concept sensitivity is not uniformly ordered as BioViL-T encodes Pneumothorax strongly **(0.7350 vs 0.5900)** while CXR-CLIP encodes Cardiomegaly strongly **(0.6100 vs 0.5750)**, with Pleural Effusion tied. The similarity TCAV further reveals that BioViL-T's image and text encoders are not uniformly coupled across pathologies **(Cardiomegaly 0.7100, Pneumothorax 0.4700)**, indicating that each architecture encodes different pathologies with different strengths.

### 5.3 Faithfulness [F]

Faithfulness asks whether the regions a heatmap highlights actually drive the model's predictions. Three measurements are reported per heatmap; deletion AUC, insertion AUC, and the saliency-versus-random mask drop. A faithful heatmap should produce a low deletion AUC, a high insertion AUC, and a saliency drop that exceeds the random drop.

Table 3: Faithfulness Comparison: BioViL-T vs CXR-CLIP

| Metric | BioViL-T | CXR-CLIP |
|---|---|---|
| Deletion AUC ↓ | 0.0281 | -0.0271 |
| Insertion AUC ↑ | 0.4163 | 0.1108 |
| Saliency Mask Drop ↑ | 0.6023 | 0.0757 |
| Random Mask Drop | 0.8121 | 0.0472 |
| Saliency Blur Drop ↑ | 0.4989 | 0.0257 |
| Random Blur Drop | 0.0649 | 0.0012 |
| Mask > Random % | 0.0667 | 0.6000 |
| Blur > Random % | 0.8667 | 0.5333 |

#### 5.3.1 Deletion and insertion

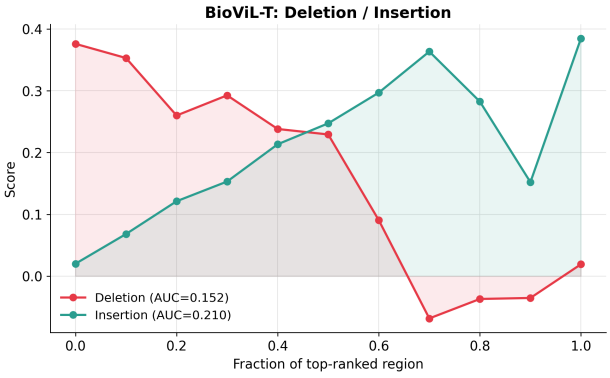



Figure 7: BioViL-T: Deletion / Insertion curves.

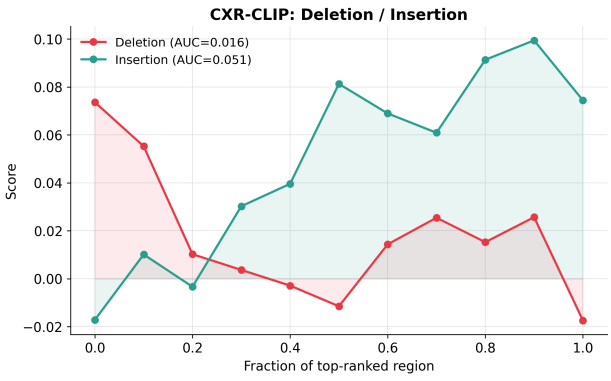



Figure 8: CXR-CLIP: Deletion / Insertion curves.

The deletion-insertion curves **(Figures 7 and 8)** tells consistent stories within each model, with a clear gap between deletion and insertion AUCs in both cases. BioViL-T's deletion curve falls from **0.375** at 0% to near zero by 70%, with the insertion curve from **0.02** to **0.38**; the deletion AUC is **0.152** and the insertion AUC **0.210**. CXR-CLIP shows the same qualitative pattern at a much smaller scale: deletion AUC **0.016**, insertion AUC **0.051**. In both models the insertion curve dominates the deletion curves, satisfying the faithfulness condition.

The absolute scale difference between the two models is large. BioViL-T's scorer is the cosine similarity between the L2-normalised global image embeddings and report embeddings. CXR-CLIP's scorer is the calibrated pos - neg prompt margin, which produces values typically below 0.1 because the positive and negative prompt directions are intentionally close in the latent space. The ten-fold scale difference between the two AUCs reflects scorer geometry and not a difference in faithfulness. A scale aware comparison normalises the insertion-deletion gap by each model's unperturbed score. BioViL-T's gap of **0.058** represents 15% of its typical **0.375** score, while CXR-CLIP's gap of **0.035** represents 47% of its **0.074** unperturbed score. Under this normalisation, CXR-CLIP shows a proportionally larger separation between its deletion and insertion curves than BioViL-T, even though BioViL-T's curves span a wider absolute range. The deletion-insertion test therefore credits both models as faithful.

### 5.3.2 Masking and Blurring Comparisons

The masking comparison applies a saliency-guided mask covering the top 20% of ranked pixels and contrasts the resulting score drop against the mean drop from ten random masks of the same area. A faithful heatmap should produce a saliency-guided drop exceeding the random drop in both modes.

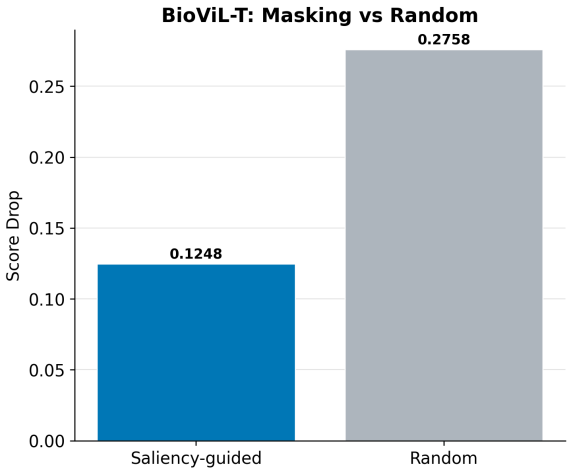



Figure 9: BioViL-T: Saliency masking vs random masking.

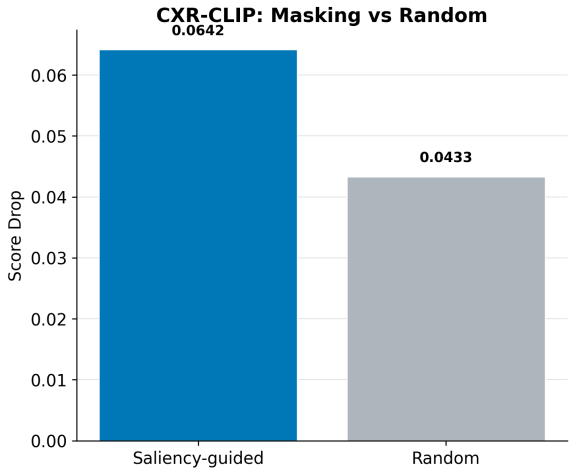



Figure 10: CXR-CLIP: Saliency masking vs random masking.

**Masking vs Random** Under hard masking, the two models diverge sharply. BioViL-T's saliency-guided drop **(0.1248)** is smaller than its random **(0.2758)**, and the per-sample win-rate Mask Random % collapses to 6.7%. CXR-CLIP shows the opposite: saliency drop **0.0642 vs random 0.0433**, with a win-rate of **60.0%**. The BioViL-T inversion is methodological rather than a faithfulness failure. Hard zero-masking introduces high-frequency edges at the mask boundary, and ten randomly scattered masks produce a larger total perimeter; hence, BioViL-T's cosine similarity scorer reacts strongly to these synthetic edges. CXR-CLIP's prompt margin scorer is bounded near zero and consequently less reactive to edges.

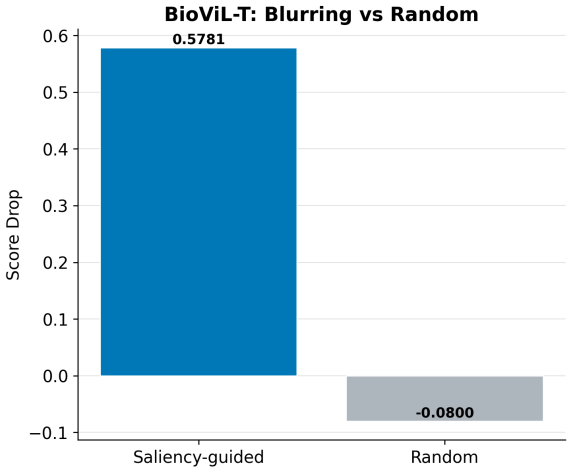



Figure 11: BioViL-T Blurring

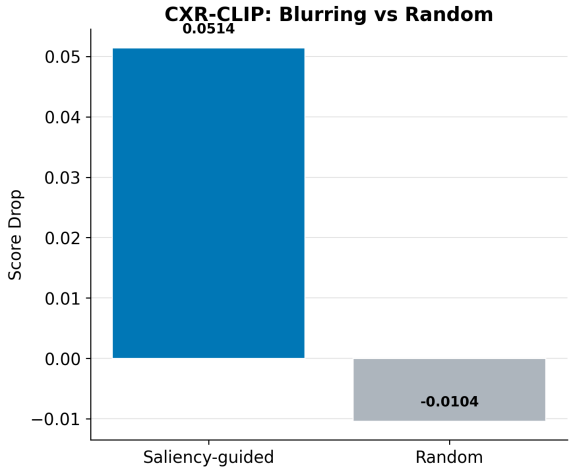



Figure 12: CXR-CLIP: Blurring

**Blurring vs Random** The blur condition removes the edge confound and reverses the picture. BioViL-T's saliency-guided drop **(0.5781)** far exceeds its random drop **(-0.0800)**, a separation of **0.6581**, with a win-rate Blur  Random % of 86.7%. CXR-CLIP shows the same direction at smaller scale: **(saliency 0.0514 vs random -0.0104)**, win-rate **53.0%**. BioViL-T's blur result is the strongest faithfulness observed in either model in either mode. CXR-CLIP's **53%** win-rate indicates that under blurring its similarity-relevance maps are indistinguishable from random, consistent with the high-entropy heatmap shape reported in Section 5.2.1.

### 5.3.3 Synthesis

The two perturbation modes converge with the deletion-insertion result of Section 5.3.1. BioViL-T's cross-modal attention concentrates on regions the model relies upon, evidenced most clearly under blurring where edge artefacts do not confound the comparison. CXR-CLIP's similarity relevance maps distribute reliance more diffusely, producing smaller but still positive faithfulness signals. Both models satisfy the faithfulness condition once scorer geometry and perturbation artefacts are accounted for.

## 5.4 Clinical Plausibility [CP]

Clinical plausibility is assessed through four artefact checks applied to 60 heatmaps per model: border_focus flags fixation on text overlays or collimation edges, rib_overfocus flags concentration on horizontal rib shapes, centre_bias flags excessively centred or de-centred activation, and activation_spread flags heatmaps that are uniformly flat or collapsed to a single pixel. A heatmap "passes" a check if it does not exhibit the artefact.

### 5.4.1 Artefact Check Comparison

Table 4: Artefact Check Summary: BioViL-T vs CXR-CLIP

| Check | BioViL-T | | CXR-CLIP | | Total |
|---|---|---|---|---|---|
| | Pass Rate | Passed | Pass Rate | Passed | |
| border_focus | 96.7% | 58 | 61.7% | 37 | 60 |
| rib_overfocus | 100.0% | 60 | 100.0% | 60 | 60 |
| centre_bias | 0.0% | 0 | 0.0% | 0 | 60 |
| activation_spread | 100.0% | 60 | 98.3% | 59 | 60 |

Three of the four checks separate the two models cleanly. BioViL-T passes border_focus on 58 of 60 heatmaps **96.7%**, while CXR-CLIP passes only 37 **61.7%**, this 35-point gap indicates that for

roughly four in ten samples CXR-CLIP places more than half of its activation within the 10% image border. This is consistent with the high-entropy heatmap shape reported in Section 5.2.1 as similarity maps spread activation across a wide region which may cause it to leak into non-diagnostic peripheral zones. BioViL-T's concentrated cross-modal alignment maps largely avoid this failure mode, with only two samples flagged. Both models pass rib_overfocus on all 60 samples, indicating that neither method fixates on horizontal edges. activation_spread is similarly clean for both models **(BioViL-T 100%, CXR-CLIP 98.3%)**, confirming that neither produces uniformly flat or single-pixel heatmaps. The centre_bias check fails universally for both models (0/60 each). This is not a faithfulness failure but a property of the check itself. The check flags heatmaps whose centre-of-mass lies either too close to the image centre or too far from it, almost all clinically relevant regions sit within the central third of the image, so anatomically reasonable heatmap will be flagged. Across both models, the centre_bias should be interpreted as a sensitivity probe rather than a pass/fail criterion.

#### 5.4.2 Synthesis

Excluding the saturated centre_bias check, BioViL-T passes 178 of 180 artefact checks (**98.9%**) versus CXR-CLIP's 156 of 180 (**86.7%**). The gap is concentrated entirely in border_focus, where CXR-CLIP's diffuse activation pattern leaks into peripheral non-clinical regions. This converges with the entropy gap in Section 5.2.1, the spatial concentration in Section 5.2, and the stronger blur-mode faithfulness in Section 5.3.2. By this we can conclude that BioViL-T's cross-modal attention produces clinically plausible focal heatmaps, while CXR-CLIP's similarity relevance maps produce less anatomically constrained activation distributions that include non-diagnostic regions.

# 6 Reflection, Future Work and Conclusion

## 6.1 Reflection

### 6.1.1 Assessment against original aims

The four aims set out in Section 2.5 are assessed individually against the evidence presented in Sections 4 and 5. The first aim requires curating two high-quality MIMIC-CXR-JPG subsets, which is fully met. The single-inference pipeline reduced 377,110 candidate rows to a balanced 10,000-sample subset spanning eight target pathologies, and the temporal-inference pipeline produced 2,000 prior–current pairs within a 1–180 day window after a deduplication strategy (Section 4.1). Both pipelines are reproducible from released code and CSV manifests.

The second aim expects this project to implement functional inference and obtain embeddings, which was satisfied with a stretch component. Both BioViL-T and CXR-CLIP pipelines run end-to-end and produce calibrated outputs; BioViL-T additionally exposes per-layer attention, per-patch projections, per-token text projections, and pre- and post-fusion temporal representations, which exceed the minimum required for the comparative study and enabled the four-quadrant temporal rollout that proved central to Section 5.

The third aim is to implement attention-, attribution-, concept-, and counterfactual-based interpretability methods. This is fully met. Eleven distinct methods are implemented across the two models, including spatial and temporal attention rollout, cross-modal alignment, Grad-CAM, Grad-CAM++, Integrated Gradients, similarity-relevance mapping, TCAV with both embedding and similarity variants, and four counterfactual families. The four-family taxonomy proposed in Section 2.4 is realised in code and exercised in the evaluation.

The fourth aim ensures fair comparison, which is also met. The same pools are balanced via `balance_pathology_sample` so that BioViL-T (single), BioViL-T (temporal), and CXR-CLIP are evaluated on identical `dicom_id` pools. CheXmask quality filtering (Dice RCA $\geq 0.7$) is applied uniformly; CXR-CLIP's calibration thresholds are fitted only on the 80% calibration split and applied unchanged to the held-out 20% test split, preventing threshold leakage.

### 6.1.2 Limitations and robustness of synthesis

The synthesis is robust to its limitations only to the degree to which the limitations are made explicit and traced through the claims they could threaten. Three limitations dominate.

CheXmask provides anatomical, not pathology-specific, masks. This biases the Pointing Game toward pathologies whose proxy regions cover large central areas (Cardiomegaly) and against peripheral lesions (Pneumothorax), inflating CXR-CLIP's overall LQ score of 0.3501. The bias does not threaten the entropy gap (Section 5.2.1), the 35-point border-focus gap (Section 5.4.1), the blur-mode faithfulness reversal (Section 5.3.2), or the TCAV asymmetry (Section 5.2.2), as none of these depends on anatomical localisation ground truth. The synthesis therefore survives this limitation; only LQ ranking is contingent on it.

CXR-CLIP is evlauated zero-shot with calibrated thresholds, while BioViL-T uses cosine similarity to th report directly. Both models see identical pools, but the scorer geometries differ in scale by an order of magnitude (Section 5.3.1). Scale-sensitive comparisons are normalised by each model's unperturbed score, and direction-of-effect statements (insertion > deletion, saliency > random) are reported alongside absolute values; the synthesis avoids ranking the models by raw AUC.

All evaluation is based on the MIMIC-CXR-JPG, Whether the same architectural patterns hold on CheXpert, NIH ChestX-ray14, or VinDr-CXR is unknown. The findings most likely to generalise are architecture-driven (the entropy gap, the gradient-vs-attention split, the border-focus failure mode); those most likely to be dataset-specific are pathology-level (the Cardiomegaly–Pneumothorax inversion may shift with prevalence and acquisition characteristics).

### 6.1.3 Methodological Lessons

Three observations emerge from the implementation and results that have implications beyond this dissertation.

Single-metric interpretability comparisons are systematically biased. CXR-CLIP's leading LQ score of 0.3501 is reframes once SA exposes its 10.69 entropy, F shows its proportionally smaller insertion-deletion gap relative to scorer scale, and CP attributes 38.3% of its samples to border focus. Any evaluation resting on the the Localisation Quality alone would have concluded that CXR-CLIP is more interpretable. a conclusion four-metric framework refutes. Future interpretability comparisons should triangulate across method families rather that an over all single localisation score.

Perturbation-based faithfulness is confounded by scorer geometry and edge artefacts. BioViL-T's masking-vs-random win-rate of 6.7% would be reported as a faithfulness, failure under a less careful protocol. The blur-mode reversal (86.7%) and the deletion-insertion monotonicity together diagnose the issue as cosine-similarity relativity to high-frequency mask boundaries, compounded by the larger total perimeter of ten scattered random masks compared with one contiguous saliency mask. Faithfulness studies on cosine-similarity-based VLMs should treat hard zero-masking as suspect by default and report blur and zero modes side-by-side.

Architecture shapes the explanations available, not just their quality. BioViL-T's cross-modal alignment and four-quadrant temporal-rollout exist only because the architecture exposes per-token projection and per-quadrant temporal attention; CXR-CLIP's ResNet-50 backbone admits Grad-CAM and similarity-relevance mapping but lack token-level and temporal explanations. The interpretability toolkit available to a model is therefore a function of architecture as much as of method, and architectural choices should be recognised as interpretability decisions.

## 6.2 Future Work

This project can be extended by incorporating Lesion-level ground truth instead of using CheXmask anatomical proxies. This can be done by replacing CheXmask from CheXlocalize or RSNA Pneumonia detections would resolve the proxy-bias in LQ and enable per-lesion rather than per-region pointing scores, isolating disease-specific localisation from anatomical overlap. This project can also be set up to imitate a real-world simulation by pairing it with a small radiologist evaluation (10-20 cases per

pathology under blinded heatmap rating) would validate CP and SA against clinical judgement and quantify the gap between automated artefact checks and clinical utility.

The four-quadrant rollout decomposition (section 4.3.1) show BioVil-T's temporal module aids Pleural Effusion localisation but not Cardiomegaly or Pneumothorax. Synthetic temporal-pair experiments pairing a current image with progressively perturbed priors would isolate whether the module attends to genuine inter-study change or to spurious co-variates such as positioning and exposure. Method ensembling can also extend this projects scope by observing complementary failure modes (gradient methods strong on central pathologies, attention rollouts on peripheral) suggesting a weighted ensemble that could outperform either famliy of interpretability methods alone. A learned combiner over the four-family methods, evaluated under the same four-metric framework, is a natural next step.

Two longer-horizon directions follow. Extending the framework to additional medical VLMs (CheXagent, RAD-DINO, MedCLIP) would test whether the architecture-driven findings (entropy gap, border-focus failure, gradient-vs-attention split) generalise beyond the BioViL-T/CXR-CLIP pair. Mitigating CXR-CLIP's border-focus failure through attention-based pooling, anatomical priors at the projection layer, or post-hoc spatial constraints would test whether the failure is intrinsic to global-pooled ResNet backbones or remediable through architectural intervention.

### 6.3 Conclusion

This project addressed a gap in medical VLM interpretability: no prior work had systematically compared multiple complementary interpretability methods across architecturally distinct VLMs in a clinical radiology setting. The contribution is fourfold. Two reproducible MIMIC-CXR-JPG subsets were curated (10,000 single-inference, 2,000 temporal pairs) under explicit quality, view, and label constraints. End-to-end inference pipelines were implemented for BioViL-T and CXR-CLIP, exposing all intermediate representations required for downstream interpretability analysis. Eleven interpretability methods were implemented spanning attention-, attribution-, concept-, and counterfactual-based families, with two methods (cross-modal alignment, four-quadrant temporal rollout) requiring architecture-specific design. A four-metric evaluation framework; Localisation Quality, Semantic Alignment, Faithfulness, Clinical Plausibility was developed under which the methods are systematically compared.

The synthesis reveals that headline localisation scores conceal architecture-driven differences. CXR-CLIP's leading LQ rests partly on centre-bias against anatomical proxies, while BioViL-T produces lower-headline but more concentrated, clinically plausible cross-modal heatmaps (entropy 5.19 vs 10.69; border-focus pass-rate 96.7% vs 61.7%). Faithfulness, once corrected for scorer geometry and edge artefacts, credits both models, with BioViL-T showing the strongest blur-mode separation observed (0.6581). Concept sensitivity is pathology-asymmetric rather than uniformly ordered, and BioViL-T's temporal module aids peripheral-pathology localisation specifically. The four-metric framework was essential to these conclusions: any single metric, applied alone, would have produced a misleading ranking.

Beyond the specific findings, this project demonstrates that interpretability comparisons in medical VLMs require multi-family triangulation, scorer-geometry-aware perturbation protocols, and explicit recognition that architecture itself is an interpretability choice. These principles transfer to any high-stakes clinical AI system where the question is not whether a model performs well, but whether the explanations it offers can be trusted.

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